Straight-line residual inspection
Mathematical and implementation guide, v1

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This guide describes the completed display algorithm derived from the owner-approved design `docs/developer/design/linearity-residual-display-v1.md`. It operates on saved ROI means, not individual image pixels. It produces exploratory diagnostics, not physical shutter or amplifier calibrations.

1 Inputs and selected population

For one detector and ROI, each individual flat exposure supplies exposure time t_i in seconds and saved bias-subtracted mean signal S_i in ADU. Both members of each pair remain separate. Their common accepted pixel population, source asset/HDU, acquisition index and original table row remain attached. There is no overscan correction or additional flat-fielding in this display. The reference-line population I contains finite (t_i, S_i) and satisfies the user's inclusive original-signal interval. Blank limits mean all available data. No weighting, outlier rejection or hidden exposure cut is applied. Points outside I remain visible and receive predictions if a line is available.

2 Centered ordinary least squares

With $n = |I|$, define

$$\bar{t} = \frac{1}{n}\sum_{i\in I} t_i, \qquad \bar{S} = \frac{1}{n}\sum_{i\in I} S_i,$$

$$b = \frac{\sum_{i\in I}(t_i - \bar{t})(S_i - \bar{S})}{\sum_{i\in I}(t_i - \bar{t})^2}, \qquad a = \bar{S} - b\bar{t}.$$

The slope b has units ADU/s and a has units ADU. A line requires at least two points and nonzero exposure-time spread. Invalid intervals, insufficient points, zero time spread and numerical overflow give explicit unavailable states. The implementation uses compensated sums. Means use the first sample as an origin, summing scaled differences; predictions use

$$\widehat{S}_i = \bar{S} + b(t_i - \bar{t})$$

rather than directly evaluating $a + bt_i$, reducing cancellation for large time origins. No coefficient uncertainties are inferred from these descriptive fits.

3 Residuals, rates and display

For each available original measurement, including outside-interval points,

$$r_i = S_i - \widehat{S}_i, \qquad r_{\%,i} = 100\frac{r_i}{\widehat{S}_i} \quad (\widehat{S}_i > 0), \qquad R_i = \frac{S_i}{t_i} \quad (t_i > 0).$$

Residuals in ADU remain available at nonpositive predictions, but percentage residuals do not. Zero-time count rates and unavailable signals remain null; negative measured signals are retained. Nonfinite derived values are null. Residual plots use original signal or exposure time on the horizontal axis, and percentage or ADU on the vertical axis, with a zero reference. Curves connect measurements for inspection; fitting limits affect the line, not displayed data. The per-ROI summary reports a, b, n, the actual selected time/signal ranges and status. The JSON download retains settings, fits, rows and ASDF source references. Original ASDF measurements are never rewritten.

4 What a shutter error can show

For an otherwise linear, correctly bias-subtracted response under stable light,

$$S_i = A(t_i + \delta) \quad \Longrightarrow \quad a = A\delta, \quad b = A, \quad R_i = A\left(1 + \frac{\delta}{t_i}\right).$$

A free-intercept line absorbs the constant shutter offset: residuals alone do not recover it. Its fractional count-rate effect grows as $1/t_i$, motivating the separate rate view. An additive residual bias also produces a $1/t_i$ term; lamp variation and detector nonlinearity can contribute structure. No physical shutter estimate is extracted here. Different signal intervals can yield different residual baselines even for the same underlying response.

5 Connection to implementation and validation

Arithmetic: `tools/pair-qc-residuals.js`, function `QcResiduals.analyze`, taking sample rows and limits, performs grouping, interval selection, OLS and per-row derived values without modifying its input.

Presentation: `tools/pair-qc-viewer.html` manages metric/axis/interval controls, ROI visibility, summaries, exact inspection and diagnostic JSON export.

Generation: `tools/view-pair-qc.py` embeds the dependency-free module into a standalone HTML document and freezes its source with the other inputs. The JSON alongside the HTML is the saved measurement payload; browser-generated fit results are separately downloadable.

Checks: `tools/tests/test_pair_qc_residuals.cjs` verifies analytical solutions, restricted intervals, rank failures, large time origins, null and signed data, and immutability. `tools/tests/test_pair_qc_js.cjs` exercises the actual embedded script with a minimal DOM surface; it does not verify browser painting. Existing Python viewer tests check ASDF/geometry/source packaging and preservation.